

Mandarin Humorous Homophone Recognition and Disambiguation in Automatic Speech Recognition

Sicheng Jin¹, Jinghao Chen¹, Mostafa Shahin¹, Beena Ahmed¹, Aditya Joshi¹

¹School of Electrical Engineering & Telecommunications,
University of New South Wales, Australia

z5317861@ad.unsw.edu.au, z5327748@unsw.edu.au, m.shahin@unsw.edu.au,
beena.ahmed@unsw.edu.au, z3539958@ad.unsw.edu.au

Abstract

Mandarin homophones remain a key challenge to improving current automatic speech recognition (ASR) accuracy due to the phonological equivalence. Mandarin speakers use this feature casually to implicitly convey emotions such as humour. Recent homophone-aware ASR studies have improved recognition accuracy, but intentional homophone twists in speech remain underexplored. In this paper, we identify patterns of homophone-based rhetorical wordplay in Mandarin, referred to as *HumourPhone*, and propose an ASR Adapter for homophone and HumourPhone recovery. Experimental results show that the proposed approach improves the recall of recognising intended HumourPhone substitutions by over 5% and achieves a 4.35% drop in target-span character error rate for homophone correction compared to baseline. These results highlight the need for homophone-aware modelling of lexical ambiguity and rhetorical wordplay in Mandarin speech recognition.

Index Terms: speech recognition, Mandarin HumourPhone, computational paralinguistics

1. Introduction

Mandarin ASR systems and speech LLMs have achieved strong performance across a wide range of speech understanding tasks. However, they still struggle in homophone-rich scenarios, where the same or similar pronunciations map to semantically distinct written forms [1]. This challenge is amplified for rare, context-dependent, or deliberately playful forms, as models tend to output high-frequency lexical choices [2].

Mandarin Chinese has a high density of homophones due to its tonal system and constrained syllable inventory; over 62.2% of Chinese characters have been reported to exhibit homophonic features [1]. Speakers often exploit this ambiguity for rhetorical wordplay and humour, including homophonic puns, allegorical sayings, and online linguistic creativity [3, 4, 5]. Such wordplay may involve identical homophones with the same syllable and tone, or near homophones with small phonetic shifts [6]. This makes Mandarin Homophone challenging for speech models, as current speech LLMs still struggle to capture subtle prosodic and pitch cues that interact with syntax, semantics, and pragmatics [7]. In this work, *HumourPhone* refers to utterances that evoke a familiar word or scenario through phonetic ambiguity and introduce an unexpected semantic twist [8, 9].

Across languages, prior work has addressed homophones and phonetic ambiguity by making pronunciation-related ambiguity explicit in training, decoding, or evaluation. In Mandarin ASR, end-to-end systems use pronunciation-aware character encodings or homophone-based label smoothing to reduce syllable-to-character confusion [10, 11]. Contextual biasing has further introduced confidence-based homophone detectors

to correct low-confidence ASR outputs in Mandarin [12]. In low-resource Cantonese ASR, homophone extension and unified writing inject homophone lexicons into beam-search decoding and normalise character variants for rare-word recognition [13]. In English, audio-pun research decomposes the understanding of spoken wordplay into staged tasks, such as pun recognition and meaning inference, to benchmark large audio language models (ALM) [14]. Together, these studies suggest a common pattern: homophone-related speech tasks benefit from explicitly linking pronunciation with candidate written forms and contextual interpretation.

Recent Mandarin homophone studies provide further evidence for this pronunciation–orthography–context connection. Beyond phonological ambiguity, homophones have been studied as intentional devices in internet slang, memes, humour, and toxicity evasion [15, 16]. Existing methods therefore often combine phonological cues with higher-level interpretation, including pinyin-guided reasoning [17], multimodal chain-of-thought (CoT) verification for homophone and pun understanding [18], and template-based or sequence-to-sequence generation supported by human-engineered lexicons [19]. However, these methods generally rely on auxiliary textual, phonological, or lexicon-based information and are mainly designed for homophone disambiguation rather than speech-based humourphone, where the intended written form and humorous semantic twist must be inferred directly from audio.

In this work, we focus on speech-based Mandarin humourphone recognition and correction, which aims to identify intentional homophone-based wordplay from speech and restore its intended written form. Ordinary homophone correction is also reported as a secondary evaluation setting. We propose a task-conditioned ASR adaptation framework that detects potential humourphone spans, generates LLM-based candidates, and selects contextually appropriate corrections. Our contributions are: 1) a humourphone-oriented ASR adapter that also supports ordinary homophone correction; 2) a target-aware evaluation protocol for measuring recognition and correction within annotated regions; and 3) a humourphone dataset for evaluating speech models on homophone-based wordplay.

2. Methodology

Fig. 1 gives an overview of the proposed task-conditioned adapter for Mandarin homophone correction and humourphone recovery. We define homophone as two words that share similar or identical pronunciation but different meanings, and humourphone as homophones intending to possess a humorous implication. Given an input utterance, the system first obtains an initial ASR transcription and then selects either the ordinary homophone branch or the humourphone branch according to the emo-

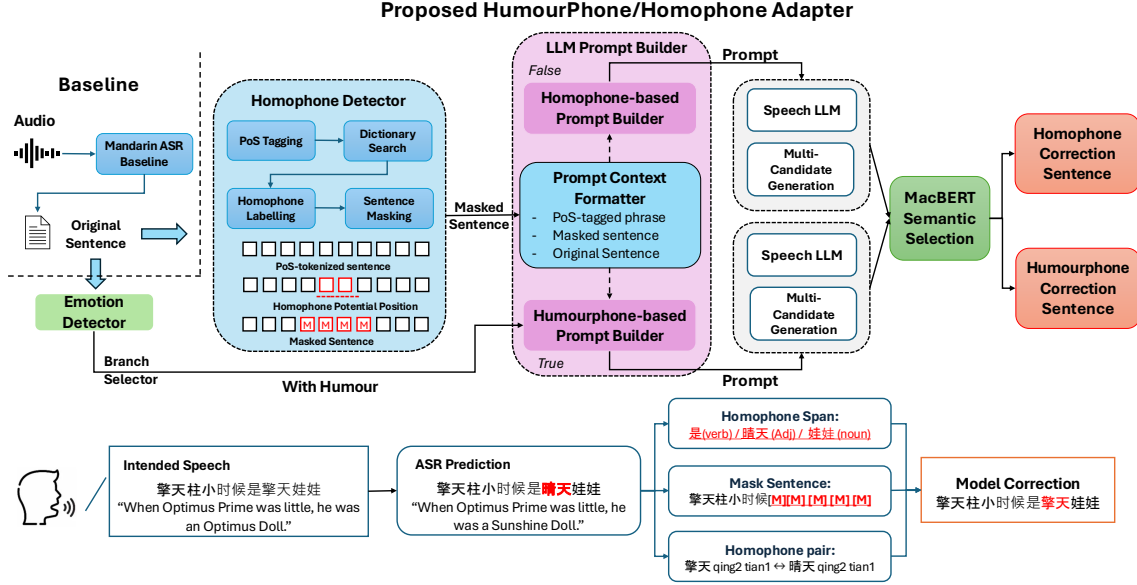


Figure 1: Workflow of the proposed adapter for potential-span-based homophone correction and humourphone recovery. In the bottom diagram, intended speech refers to the input audio, ASR prediction refers to the original sentence (i.e. baseline), while the three subsequent blocks are the output of the Homophone Detector.

tion detector output. The ordinary homophone branch focuses on detecting and correcting candidate homophone spans, while the humourphone branch incorporates humour-oriented contextual cues to recover intentional homophonic wordplay. In both branches, task-specific information is converted into prompts for the speech LLM, which generates multiple candidate transcriptions from the original audio and contextual guidance. The final output is selected using a semantic selector to balance correction accuracy and the risk of overcorrection.

2.1. Humourphone Dataset Construction

We collect humourphone instances from Chinese internet sources, including both plain-text examples and images containing textual wordplay. For image-based samples, the text is manually extracted and normalised. To evaluate speech-based systems, we synthesise Mandarin audio from the collected text using Microsoft Edge TTS [20], resulting in 80 audio samples.

All samples are manually inspected before evaluation. We removed samples that involve unreliable TTS pronunciation, ambiguous polyphonic characters, inappropriate content, non-humorous cases, sarcasm-only examples, or rare derogatory expressions. The final humourphone evaluation set is categorised into two types: Full Homophones, where two words share identical pronunciation, and Near Homophones, where the humourphone involves slight phonemic alteration of the original word.

2.2. Humourphone Pattern Guidelines

Based on the retained samples and literature on Mandarin homophonic wordplay [8, 9], we identify recurring humourphone patterns to guide the emotion detector and the prompt builder in the humourphone branch. These cases typically involve a recognisable source expression, scenario, or discourse structure that is intentionally disrupted through identical or near-homophonic substitution.

We summarise the observed patterns into six prompt-

level guidelines: Phonetic Phrase Deviation, Phonetic Code-Switching, Rapid Q&A and Punchline Structure, Anachronistic Semantic Clashes, Typographical and Visual Wordplay, and Anthropomorphic Attribution. These guidelines cover cases where homophones modify idioms or fixed expressions, imitate foreign words, letters, or numbers, create compact setup-punchline structures, or rely on contextual, visual, or anthropomorphic incongruity. Rather than serving as strict linguistic categories, these guidelines help distinguish intentional homophonic wordplay from ordinary ASR errors.

2.3. Homophone and Humourphone Adapter

The proposed adapter consists of three modules: a Homophone Detector, a branch-based LLM prompt builder, and a BERT-based semantic output selector. Given a raw Mandarin transcription, the Homophone Detector first performs Part-of-Speech (PoS)-aware potential homophone span detection. In parallel, an Emotion Detector, using GPT-5.5 and the previously mentioned guidelines, determines if the utterance is processed by the homophone branch or the humourphone branch.

The transcription is then segmented and PoS-tagged using Jieba [21]. Candidate tokens or spans are ranked using a Chinese word-frequency dictionary [22], where low-frequency or contextually suspicious spans are prioritised. To reduce boundary-localisation errors, the top-ranked spans are expanded with neighbouring PoS-tagged phrases. The resulting masked sentences and tagged span information are then passed to the LLM prompt builder to constrain candidate generation and avoid random sentence-level rewriting.

For homophone correction, the prompt encourages local substitutions of homophones or near-homophones. For humourphone recovery, the prompt is further augmented with the humourphone pattern guidelines defined in Section 2.2. Generated candidates are post-processed and scored using a BERT-based semantic selector [23]. In the homophone branch, the most se-

manically similar candidate is selected, with the original transcription retained within a 5% relative margin to reduce over-correction. In the humourphone branch, the selector favours lower-similarity candidates that remain consistent with the detected homophonic pattern, reflecting the semantic incongruity of intentional wordplay.

3. Experiments and Results

3.1. Data

We construct Mandarin homophone evaluation sets from AISHELL-3 (AS-3) [24] and CommonVoice 25 Chinese (CV25-CN) [25]. Homophone targets are identified using the ChineseHomophones dictionary [26], which groups expressions by pinyin sequence. The final CN Dataset is formed by merging the AS-3 and CV25-CN subsets using unique pinyin-based homophone categories, with invalid targets such as person and location names removed. HumourPhone is excluded from this merged set due to its small scale and is evaluated separately for humour-oriented analysis. Table 1 summarises the resulting datasets.

Table 1: *Statistics of the evaluation datasets. The CN Dataset is merged from AS-3 and CV25-CN after filtering invalid target entries. HumourPhone is newly constructed in this work for humour-oriented evaluation. Hom. Groups and Hom. Occ. denote homophone categories and target occurrences.*

Dataset	Hom. Groups	Hom. Occ.	Hours
CN Dataset (with filter)	1,303	85,706	125.73
HumourPhone (proposed)	80	80	0.08

3.2. Experimental Setup

We evaluate Mandarin homophone ASR performance on the merged AS-3+CV25-CN set in an inference-only setting, without additional training or fine-tuning. We compare three settings: standard ASR baselines, including Qwen3-ASR-1.7B in ASR mode (*Qwen3-ASR*) [27] and Whisper-large-v3 (*Whisper-v3*) [28]; a speech-LLM decoding baseline using Qwen3-ASR-1.7B with a basic task prompt (*Qwen3-LLM*); and the proposed emotion-guided Homophone and HumourPhone adapter.

For frequency-based analysis, target forms are labelled as *common* or *uncommon* according to corpus frequency and cross-checked with a word-frequency dictionary [22]. The adapter uses Jieba [21], Qwen2.5-7B-Instruct [29] for candidate generation, and Chinese MacBERT-large [23] for semantic selection. The emotion detector is evaluated separately on HumourPhone. Using our guidelines, it reached an accuracy in predicting raw transcripts of Qwen3-ASR and Whisper-Large-v3 at 97.0% and 76.0%, respectively. All references and outputs are normalised before scoring by converting Traditional Chinese to Simplified Chinese and expanding numerals into Mandarin written forms.

3.3. Evaluation Metrics

We evaluate Mandarin homophone ASR using target-span character error rate (T-CER), computed by character-level Levenshtein distance [30] over annotated target spans only. For correction, we report Recovery Rate (RR), the proportion of initially incorrect targets corrected by the pipeline, and Detection

Rate (DetR), which measures whether an annotated target span overlaps with a predicted mask span.

For HumourPhone, we use T-CER and Mean Uncommon Recall (MUR), which measure recall over uncommon HumourPhone characters in the target span. We also report MacBERT-based Utterance-level Semantic Similarity (USS) and Target-span Semantic Similarity (TSS) to assess semantic consistency at full-utterance and target-span levels.

3.4. Homophone Differentiation

Table 2 reports the effect of the proposed adapter on Whisper-v3 and Qwen3-ASR across common and uncommon homophone targets. We also evaluate a prompt-only speech-LLM baseline, Qwen3-LLM, but omit it from the table to focus on the adapter comparison. This baseline does not improve over Qwen3-ASR, with overall T-CER increasing slightly from 3.60% to 3.70%. This suggests that instruction-only speech-LLM decoding is insufficient for fine-grained homophone correction, as the model may struggle to locate suspicious homophone spans and can introduce incorrect substitutions.

With the proposed adapter, Whisper-v3 improves on both frequency groups. T-CER decreases from 6.43% to 5.07% on common forms, and more substantially from 24.74% to 20.39% on uncommon forms. The larger 4.35% drop on uncommon homophones suggests that the adapter is particularly useful for correcting less frequent lexical choices. This is also reflected in the higher RR of Whisper-v3+Adapter, since Whisper-v3 produces more unstable lexical realisations and occasional phrase-level errors in Mandarin transcription, leaving more recoverable errors for span-level correction.

For Qwen3-ASR, the adapter gives a smaller improvement on uncommon forms, reducing T-CER from 13.56% to 12.85%. However, T-CER on common forms increases from 2.19% to 3.42%, suggesting that the stronger Mandarin ASR baseline already resolves many frequent homophones correctly, and that the adapter may over-correct reliable outputs in some cases. Qualitative inspection shows that many of the remaining Qwen3-ASR errors involve highly confusable lexical variants, such as the alternative written forms “部份” and “部分”. Overall, these results suggest that the adapter is more beneficial for weaker or less stable ASR outputs, while stronger ASR systems require more conservative correction strategies.

Table 2: *Baseline (Base) and adapter-enhanced (Adapt.) ASR performance on homophone correction in the CN Dataset. T-CER, DetR, and RR denote target character error rate, detection rate, and recovery rate, respectively. Common/uncommon refer to high-/low-frequency reference forms based on corpus frequency and dictionary. All metrics are percentages.*

Model	Type	Occ.	T-CER ↓		DetR ↑	RR ↑
			Base	Adapt.		
Whisper-v3	Common	75,079	6.43	5.07	37.22	7.18
	Uncommon	10,627	24.74	20.39	47.27	20.87
Qwen3-ASR	Common	75,079	2.19	3.42	44.42	2.79
	Uncommon	10,627	13.56	12.85	50.67	8.94

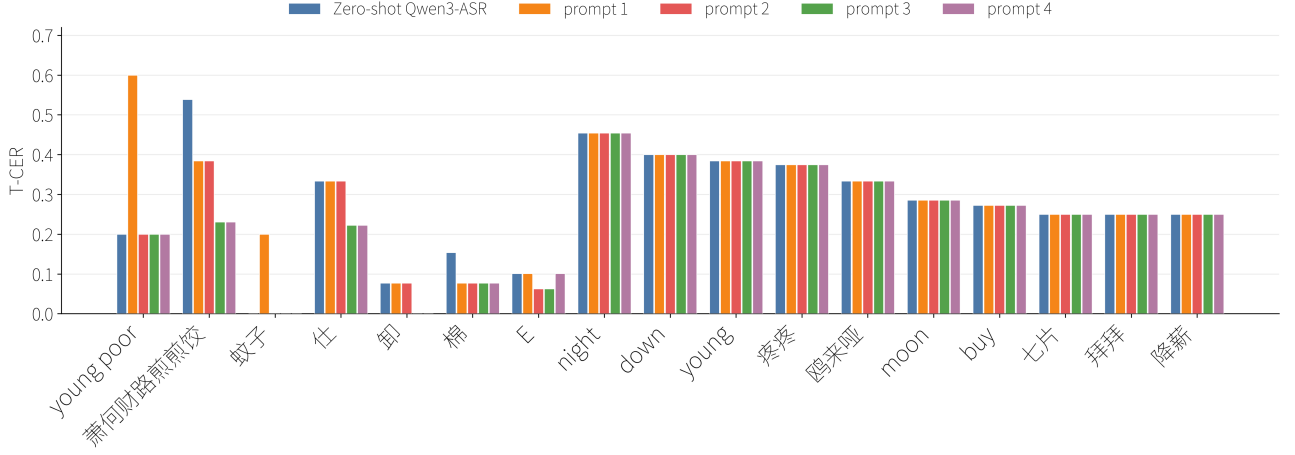


Figure 2: T-CER of different humourphone words transcribed by Qwen3-ASR across prompt variants. Some prompts reached T-CER=0 for certain words, hence the missing bars. Lower T-CER is better.

Table 3: Comparison of predictions from ASR prompt variants for non-full-homophone, full-homophone, and overall humourphone cases. The Utterance Semantic Similarity (USS) and Target-span Semantic Similarity (TSS) are calculated from the MacBERT Semantic Selector in Figure 1. We compare the candidates produced by the Speech LLM with prompts constructed with the Homophone Detector as shown in Figure 1. Lower T-CER is better; higher Mean Uncommon Recall, USS and TSS are better.

Method	Near-homophone				Full-homophone				Overall			
	T-CER ↓	MUR ↑	USS ↑	TSS ↑	T-CER ↓	MUR ↑	USS ↑	TSS ↑	T-CER ↓	MUR ↑	USS ↑	TSS ↑
Qwen3-ASR	11.61	0.6116	0.9902	0.9793	16.96	0.3095	0.9814	0.9614	15.21	0.4043	0.9854	0.9696
1-English-basic	13.15	0.6295	0.9924	0.9818	17.33	0.2810	0.9796	0.9596	15.97	0.3903	0.9855	0.9697
2-Chinese-basic	10.65	0.6295	0.9904	0.9798	16.50	0.3381	0.9814	0.9614	14.59	0.4295	0.9856	0.9697
3-Chinese-guidelines	9.69	0.6473	0.9931	0.9823	15.92	0.3667	0.9812	0.9629	13.88	0.4547	0.9867	0.9718
4-Chinese-guidelines with example	9.69	0.6473	0.9931	0.9823	16.15	0.3381	0.9814	0.9596	14.04	0.4351	0.9868	0.9700

3.5. HumourPhone Contextual Adaptation

We further evaluate Qwen3-ASR [27] on the constructed HumourPhone dataset, which contains humorous Mandarin utterances involving humour-bearing homophones. Unlike homophone evaluation, the target word may be intentionally ambiguous or semantically unexpected due to wordplay. This experiment examines whether ASR can recover such humour-bearing target spans under different levels of contextual guidance. As shown in Table 3, we compare a zero-shot Qwen3-ASR [27] baseline with four prompt settings. In the zero-shot setting, the audio is transcribed without any task-specific instruction. For Prompts 1–4, we progressively modify the instructions used by the HumourPhone-based prompt builder, following the guidelines in Section 2.2. Initially, the context was provided only in English; for subsequent prompts, we changed the instruction to Chinese. At prompt 3, all the aforementioned guidelines are incorporated, and an example is provided for prompt 4. We evaluate all settings using the metrics defined in Section 3.3.

The results show that contextual prompting can improve the model’s ability to disambiguate intentional wordplay, although the effect depends on the prompt design. Prompt 1 slightly degrades performance relative to the zero-shot baseline, increasing the overall mean T-CER by 0.76%, from 15.21% to 15.97%. However, with richer contextual background, this trend reverses starting with prompt 2. Prompt 3 achieves the strongest overall performance, reducing mean T-CER by 8.7% relative, from 15.21% to 13.88%. The improvement is more pronounced for near-homophones, where T-CER decreases by 16.5% relative,

from 11.61% to 9.69%. Near-homophone TSS also increases by an absolute 0.003, reaching 0.9823. However, prompt 4 slightly worsened the overall performance. These results suggest that targeted contextual information generally helps the model select the intended, less frequent humourphone characters rather than defaulting to common homophones.

However, as illustrated in Figure 2, the gains remain limited, especially for full homophones. The difference in gains is not associated with whether the humourphone substitution is fully Chinese or bilingual, or the length of the words. For example, the leftmost phrase “young poor” is a humourphone of the Chinese word “杨浦”, which had a lower T-CER than the full Chinese pair “仕” and “事”. Since full homophones share identical syllables and tones, the model cannot rely on phonetic deviation and must instead infer the intended humorous form from contextual and pragmatic cues. This suggests that prompting alone is insufficient for robust HumourPhone recovery. Future work should incorporate more HumourPhone training examples and confidence-aware correction strategies to reduce over-correction of intentional wordplay.

4. Conclusion

This paper proposed a task-conditioned ASR adapter to tackle homophone recognition and humourphone recovery. Experiments show that standard ASR systems are biased toward common homophone forms, whereas simple LLM decoding with a basic prompt yields only limited improvement. Adapter-

based post-processing and targeted prompting show promising but limited results for both ordinary homophone correction and humourphone recovery. Future work will expand the humourphone dataset and develop confidence-aware routing to reduce overcorrection on reliable Mandarin ASR outputs.

5. References

- [1] Y. Sun, Y. Qin, W. Chen, X. Li, and C. Li, “Context-aware multimodal fusion with sensor-augmented cross-modal learning: The blaf architecture for robust chinese homophone disambiguation in dynamic environments,” *Applied Sciences*, vol. 15, no. 13, p. 7068, 2025.
- [2] L. Fu, Y. Xin, S. Zeng, L. Fan, Y. Wu, and X. He, “Pac: Pronunciation-aware contextualized large language model-based automatic speech recognition,” in *ICASSP 2026-2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2026, pp. 16 377–16 381.
- [3] R. Xu, “Exploring chinese humor generation: A study on two-part allegorical sayings,” in *2024 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2024, pp. 1–8.
- [4] A. S. Partington, “A linguistic account of wordplay: The lexical grammar of punning,” *Journal of Pragmatics*, vol. 41, no. 9, pp. 1794–1809, 2009.
- [5] D. A. Alindra, F. Y. Siregar, and D. S. P. Siregar, “Online language phenomena in modern mandarin: A linguistic study of popular expressions and digital neologisms,” *Linguistik, Terjemahan, Sastra (LINGTERSA)*, vol. 7, no. 1, pp. 59–69, 2026.
- [6] C. Hiruncharoenvate, Z. Lin, and E. Gilbert, “Algorithmically bypassing censorship on sina weibo with nondeterministic homophone substitutions,” in *Proceedings of the international AAAI conference on web and social media*, vol. 9, no. 1, 2015, pp. 150–158.
- [7] Z. Wang, X. Zhang, X. Jiang, K. Song, and J. Yu, “Can ai understand mandarin speech prosody? a framework and benchmark showcase,” in *Proc. Interspeech 2025*, 2025, pp. 5378–5382.
- [8] N. Xu, J. Sheng, C. Li, C. Zhou, Y. Li, T. Du, J. Wang, Z. Fu, J. Li, and S. Ji, “‘‘ i see what you did there’’: Can large vision-language models understand multimodal puns?” *arXiv preprint arXiv:2604.05930*, 2026.
- [9] S. Hong, “Homophones in mandarin chinese,” *International Journal of Linguistics*, vol. 16, p. 79, 12 2024.
- [10] Y. Zheng, X. Yang, and X. Dang, “Homophone-based label smoothing in end-to-end automatic speech recognition,” *arXiv preprint arXiv:2004.03437*, 2020.
- [11] P. Shen, X. Lu, and H. Kawai, “Pronunciation-aware unique character encoding for rnn transducer-based mandarin speech recognition,” in *2022 IEEE Spoken Language Technology Workshop (SLT)*. IEEE, 2023, pp. 123–129.
- [12] C. Yang, L. Zheng, S. Tian, G. Cheng, S. Xiao, and T. Li, “Contextual biasing with confidence-based homophone detector for mandarin end-to-end speech recognition,” in *INTERSPEECH*, 2024.
- [13] H. Chung, J. Li, P. Liu, W.-K. Leung, X. Wu, and H. Meng, “Improving rare words recognition through homophone extension and unified writing for low-resource cantonese speech recognition,” in *Proc. ISCSLP*, 2022, pp. 26–30.
- [14] Y. Su, S. Zhong, Y. Zhu, R. Wang, Z. Huang, Q. Wang, N. Zhao, D. Benavides-Prado, and M. Witbrock, “Words at play: Benchmarking audio pun understanding in large audio-language models,” *arXiv preprint arXiv:2603.18678*, 2026.
- [15] D. Lin, T. Lan, J. Zhu, J. Li, W. Chen, X. Liu, X. Su, H. Hou, G. Gao *et al.*, “Exploring the capability boundaries of llms in mastering of chinese chouxian language,” *arXiv preprint arXiv:2604.15841*, 2026.
- [16] H. Guo, J. He, J. Ma, H. Na, Z. Wang, H. Zhang, Q. Chen, W. Wang, Z. Shi, T. Shen *et al.*, “Lost in pronunciation: Detecting chinese offensive language disguised by phonetic cloaking replacement,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, 2025, pp. 2538–2550.
- [17] J. Ma, Z. Feng, H. Song, E. Chersoni, and Z. Chen, “Reasoning or memorization? investigating llms’ capability in restoring chinese internet homophones,” in *Proceedings of the 3rd Workshop on Towards Knowledgeable Foundation Models (KnowFM)*, 2025, pp. 120–139.

- [18] Z. Wang, Y. Rong, D. Jiang, H. Wu, S. Zhou, and B. Xu, "Cieasr: Contextual image-enhanced automatic speech recognition for improved homophone discrimination," in *Proceedings of the 32nd ACM International Conference on Multimedia*, 2024, pp. 915–924.
- [19] Y. Su, Y. Zhu, R. Wang, Z. Huang, D. Benavides-Prado, and M. Witbrock, "A survey of pun generation: Datasets, evaluations and methodologies," *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 7375–7395, 2025.
- [20] rany2, "edge-tts," <https://pypi.org/project/edge-tts/>, 2026, python package version 7.2.8.
- [21] J. Sun, "jieba: Chinese Text Segmentation," <https://github.com/fxsjy/jieba>, 2026, python Chinese word segmentation toolkit.
- [22] lxs602, "Chinese Word Frequencies," <https://github.com/lxs602/Chinese-Mandarin-Dictionaries/tree/main/Chinese%20Word%20Frequencies>, aggregated Chinese character and word frequency resource;.
- [23] Y. Cui, W. Che, T. Liu, B. Qin, S. Wang, and G. Hu, "Revisiting pre-trained models for Chinese natural language processing," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: Findings*. Online: Association for Computational Linguistics, Nov. 2020, pp. 657–668. [Online]. Available: <https://www.aclweb.org/anthology/2020.findings-emnlp.58>
- [24] Y. Shi, H. Bu, X. Xu, S. Zhang, and M. Li, "AISHELL-3: A Multi-Speaker Mandarin TTS Corpus and the Baselines," in *Proceedings of INTERSPEECH*, 2021, pp. 2756–2760. [Online]. Available: <https://arxiv.org/abs/2010.11567>
- [25] Common Voice, "Common Voice Scripted Speech 25.0 – Chinese (China)," Mozilla Data Collective, Mar. 2026, dataset ID: cmn3iaztg00e4mb070uvufz7q; License: CC0-1.0. [Online]. Available: <https://mozilladatasetcollective.com/datasets>
- [26] LiangsLi, "Chinesehomophones: Chinese homophone word and character lexicon," <https://github.com/LiangsLi/ChineseHomophones>, 2020, gitHub repository, accessed 9 June 2026.
- [27] X. Shi, X. Wang, Z. Guo, Y. Wang, P. Zhang, X. Zhang, Z. Guo, H. Hao, Y. Xi, B. Yang, J. Xu, J. Zhou, and J. Lin, "Qwen3-asr technical report," *arXiv preprint arXiv:2601.21337*, 2026. [Online]. Available: <https://arxiv.org/abs/2601.21337>
- [28] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," *arXiv preprint arXiv:2212.04356*, 2022. [Online]. Available: <https://arxiv.org/abs/2212.04356>
- [29] Q. Team, "Qwen2.5: A party of foundation models," September 2024. [Online]. Available: <https://qwenlm.github.io/blog/qwen2.5/>
- [30] V. I. Levenshtein, "Binary codes capable of correcting deletions, insertions and reversals," *Soviet Physics Doklady*, vol. 10, pp. 707–710, 1966.